

Artificial Intelligence – Assignment

AI Agent: Smart Travel Planning Agent

Name: _____ Semester: 6th

Course: Artificial Intelligence

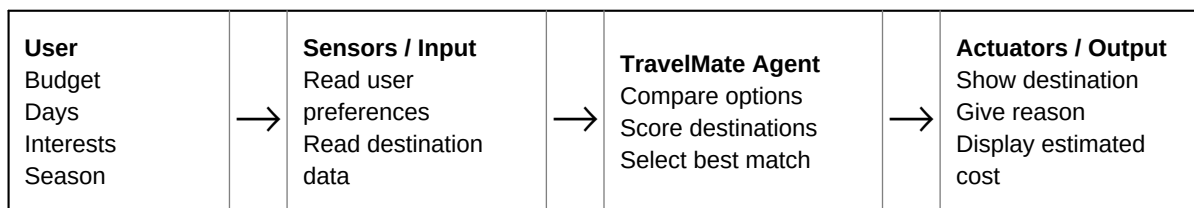
1. Think of an AI Agent

For this assignment, I selected a **Smart Travel Planning Agent**. The agent helps a user choose a suitable travel destination by looking at information such as budget, number of days, preferred type of place, and season. It then gives a simple travel suggestion. I chose this agent because it is easy to understand and it is also related to a real-life problem.

2. Name of the Agent

Name: TravelMate – Smart Travel Planning Agent

3. Graphical Representation



The basic idea is that the user gives information to the agent. The agent processes this information, checks the available destinations, and produces a travel recommendation.

4. Agent Formulation

Environment: The environment contains the user, destinations, hotels, transport options, weather/season information, and approximate travel costs. In a real system, this information can change, especially prices and availability.

Sensors: User input such as budget, number of days, preferred destination type, and season. The agent can also receive destination information such as estimated cost and activities.

Actuators: The agent displays a recommended destination, estimated cost, suggested activities, and a short explanation of why the destination was selected.

Agent Goal: To recommend a destination that matches the user's preferences while staying as close as possible to the user's budget.

5. Environment Types

Property	TravelMate Environment	Reason
Observability	Partially observable	The agent does not know everything about the user's preferences or future prices.
Determinism	Stochastic	Travel costs, weather, traffic and availability can change.
Episodic / Sequential	Sequential	The choice of destination can affect later choices such as hotel and activities.

Property	TravelMate Environment	Reason
Static / Dynamic	Dynamic	Prices and availability may change with time.
Discrete / Continuous	Mostly discrete	Destinations, preferences and choices are mostly represented as separate options.
Agents	Mostly single-agent	The program makes the recommendation, although real travel involves many outside actors.
Known / Unknown	Partly unknown	The agent may not have complete or updated information about the travel environment.

6. PEAS Framework

PEAS Part	For TravelMate
P – Performance Measure	Good destination match, stays within budget, suitable number of days, and useful recommendation.
E – Environment	Destinations, transport, hotels, activities, prices, weather/season, and the user's travel requirements.
A – Actuators	Display destination recommendation, estimated cost, activities and explanation on the screen.
S – Sensors	User inputs and available destination/travel information.

7. Short Conclusion

TravelMate is a simple goal-based AI agent idea. It takes information from the user, processes the available travel choices, and gives a recommendation. The PEAS framework helps us clearly identify what the agent is trying to achieve, where it works, how it receives information, and how it produces an action.

Note: The second part of the assignment contains a Python Travel Agent program. The code is kept simple so it can be run directly in Google Colab.